

Data-driven estimation of rare event probabilities for spatial extremes

@ISNPS 2026, Thessaloniki, Greece

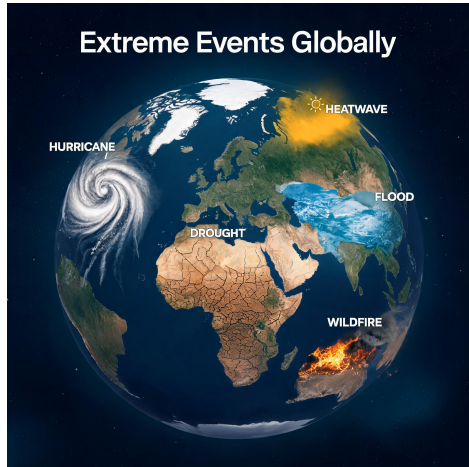


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Decision Science, Indian Institute of Management Bangalore, India

June 23, 2026

Motivation

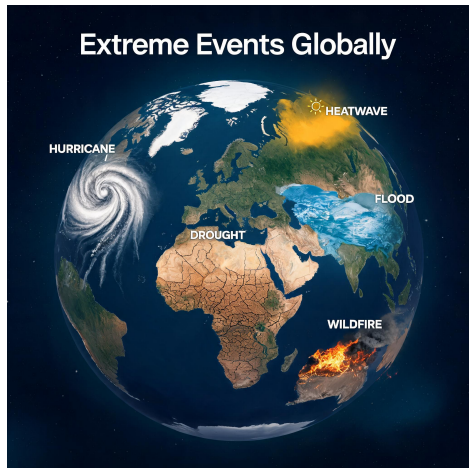
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Generated by AI

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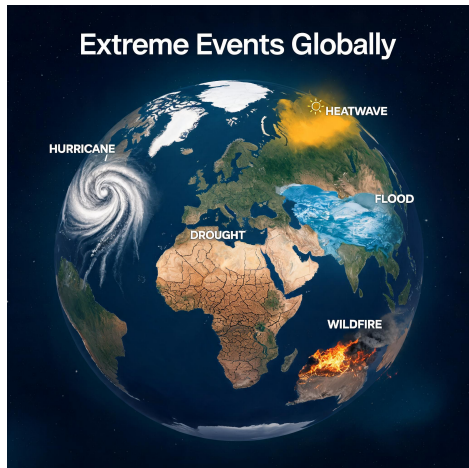
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- These events are **rare**, but their impacts are often **severe** in terms of human loss, economic disruption, environmental damage.



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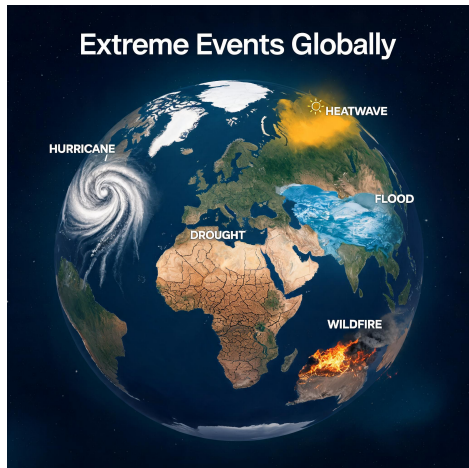
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- These events are **rare**, but their impacts are often **severe** in terms of human loss, economic disruption, environmental damage.
- A common **statistical challenge** is that such events **lie in the heavy-tailed** distribution and are **difficult to estimate** using conventional methods.



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Motivation

- **Extreme events** occur across all major regions of the world, including **hurricanes**, **floods**, **wildfires**, **heatwaves**, **droughts**
- These events are **rare**, but their impacts are often **severe** in terms of human loss, economic disruption, environmental damage.
- A common **statistical challenge** is that such events **lie in the heavy-tailed** distribution and are **difficult to estimate** using conventional methods.
- This **motivates** the need for extreme value theory (EVT) to estimate **rare event** probabilities more accurately.



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Heavy-tailed distributions

Regularly varying function

A function $f : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is said to be *regularly varying*¹ with index $\rho \in \mathbb{R}$ if for every $x > 0$,

$$\lim_{t \rightarrow \infty} \frac{f(tx)}{f(t)} = x^\rho, \quad \text{or} \quad f(tx) \sim x^\rho f(t), \quad t \rightarrow \infty$$

We denote $f \in \text{RV}(\rho)$.

Slowly varying function

For $\rho = 0$, i.e, if for every $x > 0$

$$\lim_{t \rightarrow \infty} \frac{L(tx)}{L(t)} = 1 \quad \text{or} \quad L \in \text{RV}(0).$$

- **They are connected!** $f(x) = x^\rho L(x)$

¹Bingham et al., Regular variation (1987)

²Nair et al., The Fundamentals of Heavy Tails (2022)

Heavy-tailed²

LogNormal•

• Weibull

Regularly varying

• Cauchy

• Burr

• Student's t

• L'evy

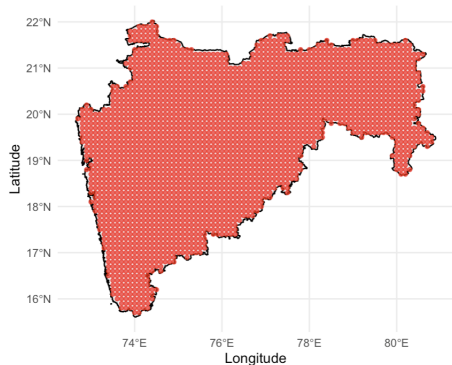
• Fr'echet **Scale invariant**

• Zipf

• Pareto

Problem Definition

- Consider a geographic region K , with data observed at discrete locations.

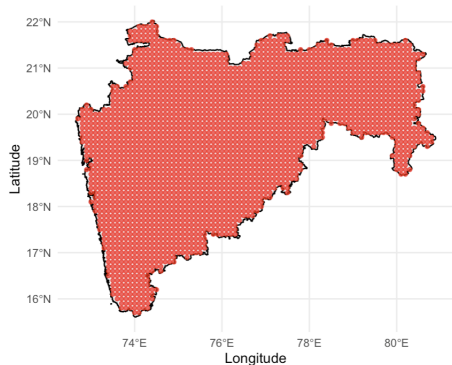


Spatial Temperature Location of Maharashtra, India
(1991-2025)

Problem Definition

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- We aim to estimate the probability that the **average regional rainfall** exceeds a critical (**extreme**) threshold t :

$$A_t = \left\{ f : \frac{1}{|K|} \int_{\mathbf{x} \in K} f(\mathbf{x}) d\mathbf{x} \geq t \right\}$$



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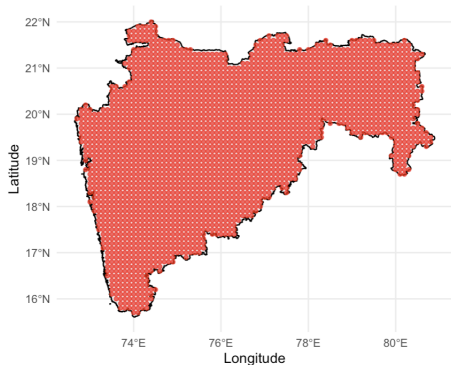
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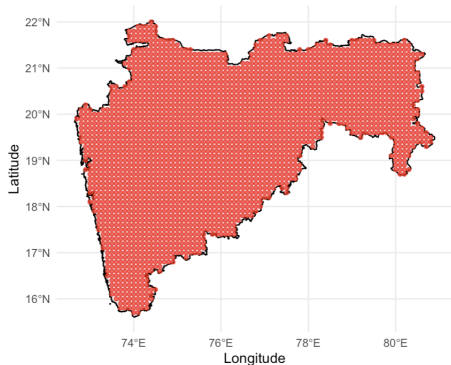
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$$A_t = \left\{ f : \sup_{\mathbf{x} \in K} f(\mathbf{x}) \geq t \right\}$$

- If t lies beyond the range of the observed data, A_t is known as a **failure set**.



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Methodology

Assumption 1

There exists a limiting process $\eta \in \mathcal{C}(K)$ and $a \in \text{RV}(\rho)$ such that³

$$[a(n)]^{-1} \max_{i=1}^n \xi_i \xrightarrow{d} \eta \quad (1)$$

1. $\xi_i \sim \xi$ are independent and identically distributed.
2. $\mathcal{C}(K)$ is the space of continuous functions on compact set K .
3. $a \in \text{RV}(\rho)$, i.e., a is regularly varying with index $\rho \in \mathbb{R}$

Assumption 2 (Homogeneity Property)

Let A_t be such that $t^{-1}A_t = A_1$ for a fixed $A_1 \in \mathcal{C}(K)$ and $\inf\{\|f\| : f \in A_1\} > 0$.

³Chapter-9 of De Haan and Ferreira: Extreme value theory: an introduction(2006)

Characterization of the distribution

Definition 1 (Limiting measure)

$$\mu(A) = \lim_{n \rightarrow \infty} \frac{1}{n} \mathbb{P}(\xi/a(n) \in A).$$

Lemma 2 (Characterization of the distribution)

Under Assumption 1,

- $\mu(tA) = t^{-1/\rho} \mu(A)$ for any $t > 0$
- $\bar{F} := 1 - F$ is regularly varying with index $-1/\rho$, where F is cdf,

$$\bar{F} \in \text{RV}(-1/\rho).$$

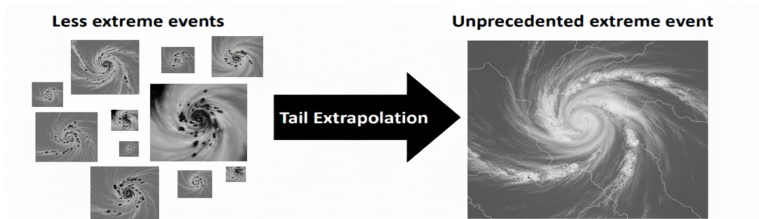
Notice: Self-Similarity of the Measure!! (the probability decays at a consistent, predictable rate (determined by ρ) regardless of how high the threshold t goes.)

Statistical estimation of probability of failure sets

- Since $\bar{F} \in \text{RV}(-1/\rho)$, $\bar{F}\left(\frac{t}{c} \cdot c\right) \sim \left(\frac{t}{c}\right)^{-1/\rho} \bar{F}(c)$ with $t \gg c$.

$$P(\xi \in A_t) \sim \left(\frac{t}{c}\right)^{-1/\rho} P(\xi \in A_c) \quad (2)$$

- $P(\xi \in A_t)$ is the **rarer probability**; ρ is the **tail index/shape parameter**
- $P(\xi \in A_c)$ is **manageable extreme level observed** in historical data
- **Practical Example:** To simulate **500mm rainfall** event (unobserved), we analyze **100mm** events (observed) with a calculated scaling index ρ .



Credit: Huser-KAUST

Questions?

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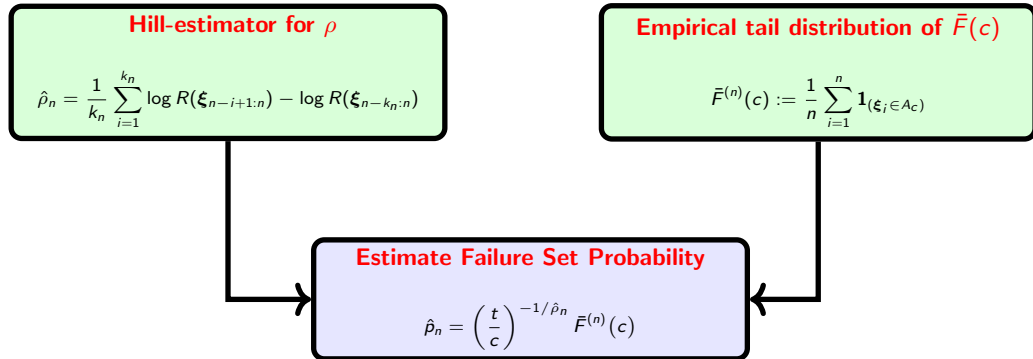
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- ✓ How do we estimate the tail distribution $P(\xi \in A_c)$ - which is $\bar{F}(c)$ at threshold c ?

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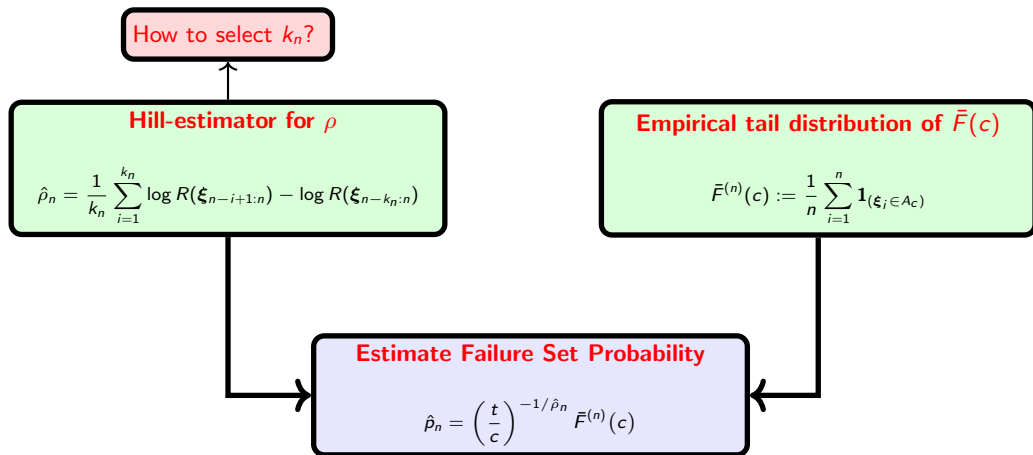
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- ✓ **How to quantify the uncertainty and the confidence intervals for the final rare-event estimate?**

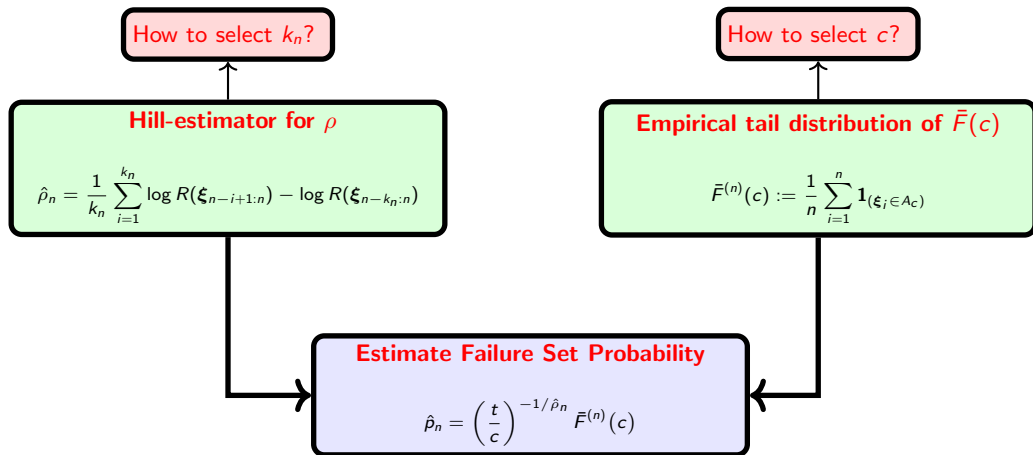
Non-parametric approach



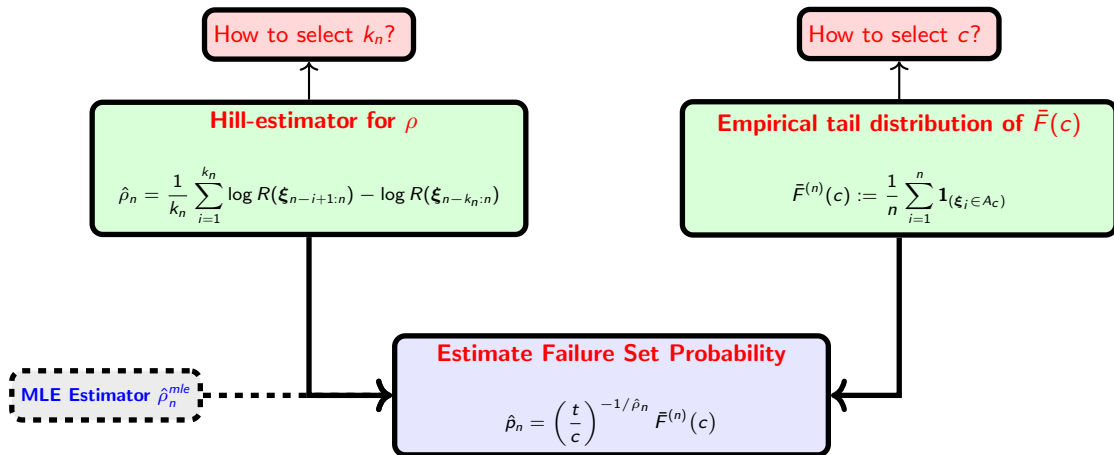
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Selection of parameter k_n

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- We choose a sequence $q_n = n^{-\nu}$, $\nu \in [\epsilon, 1 - \epsilon]$, $\epsilon \in (0, 0.5)$ such that⁵

$$k_n = \lceil nq_n \rceil = \lceil n^{1-\nu} \rceil$$

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- **Advantage:** This provides a range of k_n , so even if the chosen k_n is slightly suboptimal, the method remains robust.

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- Given $\epsilon > 0$, Plateau detected when:

$$\delta_{k_n} = |\tilde{\rho}(k_n) - \tilde{\rho}(k_n - 1)| < \epsilon,$$

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- **Advantage**: k_n^* selected **automatically**, fully data-driven.

Uniform Consistency

Uniform consistency of Hill Estimator

$$\sup_{\nu \in [\epsilon, 1-\epsilon]} |\hat{\rho}_n(k_n) - \rho| \xrightarrow{a.s.} 0$$

Uniform consistency of Empirical Tail Distribution

$$\sup_{\nu \in [\epsilon, 1-\epsilon]} \left| \frac{\bar{F}_R^{(n)}(c)}{\bar{F}_R(c)} - 1 \right| \xrightarrow{a.s.} 0$$

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Asymptotic normality

- With t chosen such that $p_n = P(\xi \in A_t) = n^{-\kappa}$ for $\kappa \in (1, T]$, $T > 1$.

Asymptotic normality

Suppose further conditions on k_n and F are satisfied, then we hope that got the following asymptotic normality

$$\frac{\rho\sqrt{k_n}}{\log d_n} \left(\frac{\hat{p}_n}{p_n} - 1 \right) \xrightarrow{d} \mathcal{N}(0, 1), \quad d_n = k_n/(np_n)$$

- In particular, we need second-order regular variation assumptions on $\bar{F}$ to control the approximation bias error.
- **Second order condition:** Suppose there exists a function $A(t)$ such that $A(t) \rightarrow 0$ as $t \rightarrow \infty$ and

$$\lim_{t \rightarrow \infty} \frac{\frac{U(tx)}{U(t)} - x^\rho}{A(t)} = x^\rho \frac{x^\psi - 1}{\psi} \text{ with } \psi < 0.$$

where $U = (1/\bar{F})^\leftarrow$ is the quantile function.

- But, we need more condition $\sqrt{k_n}A(n/k_n) \rightarrow 0, n \rightarrow \infty$

$$\frac{\rho\sqrt{k_n}}{\log d_n} \left(\frac{\hat{p}_n}{p_n} - 1 \right) \xrightarrow{d} \mathcal{N}(0, 1)$$

- With the second order condition, we can get the **speed of convergence**
- Under the above assumptions, one may get easily $\hat{d}_n/d_n \xrightarrow{P} 1$ as $n \rightarrow \infty$

$$\frac{\hat{\rho}_n\sqrt{k_n}}{\log \hat{d}_n} \left(\frac{\hat{p}_n}{p_n} - 1 \right) \xrightarrow{d} \mathcal{N}(0, 1)$$

- It establishes the **asymptotic error distribution**, enabling the construction of reliable **asymptotic confidence intervals**.
- Hence, an asymptotic $(1 - \alpha)$ confidence interval for the true p_n is

$$p_n \in \left[\frac{\hat{p}_n}{1 + z_{1-\alpha/2} \frac{\log \hat{d}_n}{\hat{\rho}_n \sqrt{k_n}}}, \frac{\hat{p}_n}{1 - z_{1-\alpha/2} \frac{\log \hat{d}_n}{\hat{\rho}_n \sqrt{k_n}}} \right].$$

where $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution.

Bootstrap Confidence Interval

- The last approximation can be demanding since, in practice, we need to know the second-order parameters.
- **Purely sample based way:** assume a second order expansion but do not assume the second order (or first order) parameters known
- Let $X_1^b, X_2^b, \dots, X_n^b$ be an i.i.d. random sample drawn from the empirical distribution function $F^{(n)}$ of the original observations $\mathcal{X}_n := \{X_1, X_2, \dots, X_n\}$

Bootstrap Confidence Interval

- With the help of an asymptotic expansion⁶

$$\sup_{x \in \mathbb{R}} \left| P(\sqrt{k_n}(\hat{\rho}_n^b - \hat{\rho}_n) \leq x \mid \mathcal{X}_n) - P(\sqrt{k_n}(\hat{\rho}_n - \rho) \leq x) \right| \rightarrow 0, \quad n \rightarrow \infty. \quad (3)$$

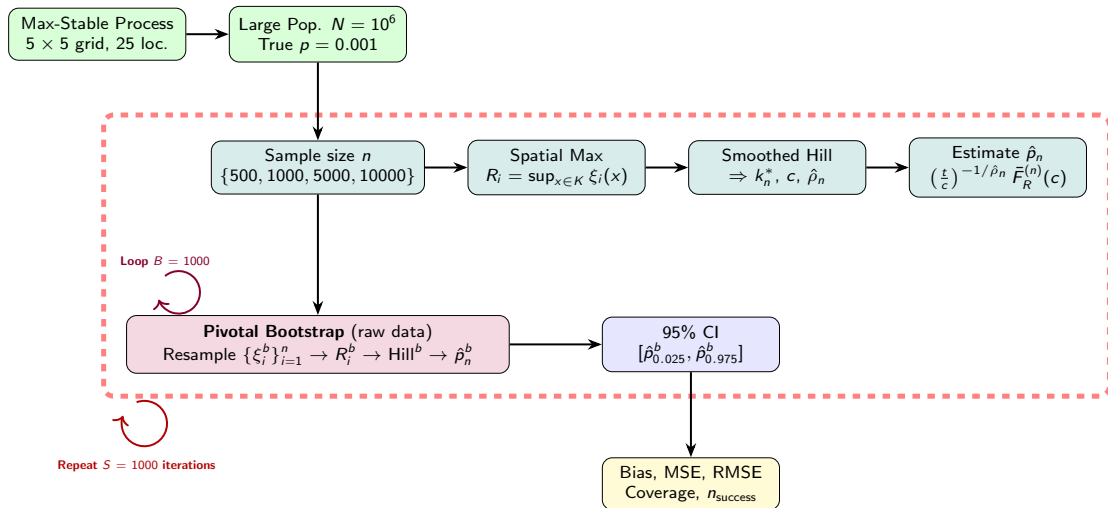
- Bootstrap pivotal $1 - \alpha$ confidence interval

$$\widehat{CI}_n := \left[2T(\hat{\rho}_n) - \zeta_{n,1-\alpha/2}, 2T(\hat{\rho}_n) - \zeta_{n,\alpha/2} \right], \quad T(\hat{\rho}_n) := \frac{\hat{\rho}_n \sqrt{k_n}}{\log \hat{d}_n} \log \hat{\rho}_n,$$

where $\zeta_{n,\alpha}$ is the bootstrap percentile quantile.

⁶de Haan & Zhou, Bootstrapping Extreme Value Estimators (2024)

Simulation Study: Design



Simulation Study

Setup

- Data generated from max-stable process using SpatialExtremes (R).
- 5×5 equally spaced grid : 25 spatial locations.
- True probability: $p = 0.001$ (99.9th percentile of 1M sample).
- Sample sizes:
 $N \in \{500, 1000, 5000, 10000\}$.
- $S = 1000$ simulation iterations.
- $B = 1000$ bootstrap replicates per iteration for CI.

Metric of assessment

- Bias : does $\hat{p}_n$ converge to p ?
- MSE / RMSE : estimation accuracy.
- Coverage : proportion of times the Bootstrap CI contain the true p
- n_{success} : how often does threshold detection succeed?

Simulation Results

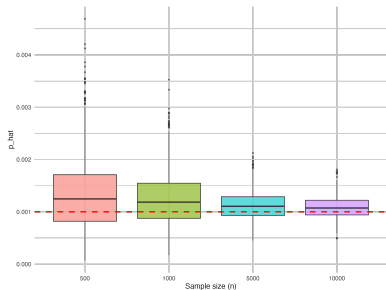


Figure: Distribution of estimated exceedance probability

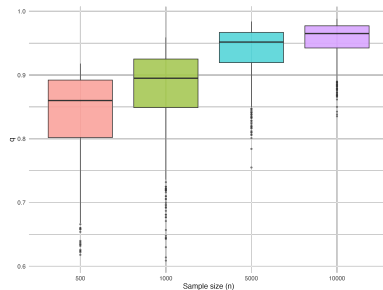


Figure: Distribution of chosen quantile level

Simulation Results: Summary Table

Table: Simulation results: $B = 1000$ iterations, $S = 1000$, true $p = 0.001$

n	$\hat{p}_n$	Bias	% Bias	MSE	RMSE	95% CI	Coverage	n_{success}
500	0.001335	0.000335	33.53	6.0×10^{-7}	0.000776	[0.000359, 0.003096]	0.968	985
1000	0.001210	0.000210	20.98	3.1×10^{-7}	0.000556	[0.000430, 0.002496]	0.968	999
5000	0.001117	0.000117	11.68	9.0×10^{-8}	0.000293	[0.000636, 0.001737]	0.971	1000
10000	0.001086	0.000086	8.56	5.0×10^{-8}	0.000220	[0.000712, 0.001542]	0.960	1000

Interpretation

- Bias decreases consistently with n : 33.5% $\rightarrow$ 8.6% : estimator is consistent.

Coverage

- Bootstrap CI achieves **96–97% coverage**

Thank You!

Madhab Barman

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Indian Institute of Management Bangalore, India

I welcome your questions and suggestions!

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